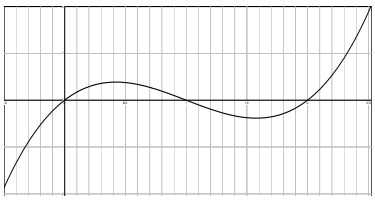
Question	Answer	Marks	Partial Marks
9(b)	$x\left(x^2-x-2x+2\right)$ or $\left(x^2-x\right)(x-2)$ or $(x-1)\left(x^2-2x\right)$ leading to x^3-3x^2+2x with no errors or omissions	**2**	**B1** for $x\left(x^2-x-2x+2\right)$ or $\left(x^2-x\right)(x-2)$ or $(x-1)\left(x^2-2x\right)$
9(c)	$3x^2-6x+2$	**B2**	**B1** for 2 correct terms
	their $\frac{dy}{dx}=0$	**M1**	
	their $\frac{-(-6)\pm\sqrt{(-6)^2-4(3)(2)}}{2(3)}$	**M2**	**M1** for $\sqrt{(-6)^2-4(3)(2)}$ or for $p=-(-6)$ and $r=2(3)$ if in form $\frac{p\pm\sqrt{q}}{r}$
	(0.4, 0.4) (1.6, –0.4)	**B3**	**B2** for 0.4 or 0.42... **and** 1.6 or 1.57 to 1.58 or for one correct pair of coordinates or **B1** for 0.4 or 0.42... **or** 1.6 or 1.57 to 1.58 If 0 scored **SC1** for $1+\sqrt{\frac{1}{3}}$ and $1-\sqrt{\frac{1}{3}}$ or better or for one correct pair of coordinates in any form
9(d)	Correct sketch 	**2**	**FT** *their* **(c)** but must be cubic i.e. correct shape cubic through origin and max and min in correct quadrants **B1** for cubic shape sketch
10(a)(i)	1	**1**	
10(a)(ii)	$\frac{1}{4}$ oe **nfww**	**2**	**M1** for $\frac{2}{4}\times\frac{2}{4}$ oe
10(a)(iii)	7	**2**	**M1** for trials with $\left(\frac{3}{4}\right)^k\times\frac{1}{4}$ soi
10(b)(i)	0.72 oe	**2**	**M1** for 0.9×0.8
10(b)(ii)	0.26 oe	**3**	**M2** for $0.9\times0.2+0.1\times0.8$ or 1 – *their* **(b)(i)** – 0.1×0.2 or **M1** for 0.9×0.2 or 0.1×0.8 or 1 – *their* **(b)(i)** or $1-0.1\times0.2$
11(a)(i)	64	**1**	

Question	Answer	Marks	Partial Marks
11(a)(ii)	127	**1**	**FT** $2 \times$ *their* **(a)(i)** – 1
11(b)	$\pm\frac{1}{2}$ oe nfww	**4**	**M1** for $(2x-1)^2+2(2x-1)$ **B1** for $4x^2-2x-2x+1$ or $(2x-1)(2x-1+2)$ **B1** for $4x^2-1$ [= 0] or $(2x-1)(2x+1)$ [= 0] OR **M1** for $x(x + 2) = 0$ (solving $g(x) = 0$) **A1** for $x = 0$ or -2 **B1** for $2x - 1 = 0$ or $2x - 1 = -2$
11(c)	$\frac{x+1}{2}$ oe final answer	**2**	**M1** for $y+1=2x$ or $\frac{y}{2}=x-\frac{1}{2}$ or $x=2y-1$
11(d)	$-\frac{1}{6}$ oe nfww	**3**	**B2** for $3x=-\frac{1}{2}$ oe OR **M1** for $2^{2x}\times 2^{x}$ oe or $4^{\frac{1}{2}x}\times 4^{x}$ oe or 8^{x} oe **M1** for $2^{-\frac{1}{2}}$ or $4^{-\frac{1}{4}}$ or $8^{-\frac{1}{6}}$ soi